Expression AI: A Real-Time De-escalation and Adaptive Response Engine for Conversational LLMs

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Github: https://github.com/Sapphirine/2026_Expression_2

Demo: https://youtu.be/QLGz5q_IB3E

Abstract

Conversational LLMs respond to the user's *current* message but are blind to the user's emotional *trajectory*. As a result, frustration tends to compound until the conversation breaks down. We present **Expression AI**, a closed-loop system that (i) annotates every turn with a 28-class GoEmotions BERT classifier, projects it into a continuous Valence/Arousal/Dominance (VAD) space, (ii) predicts whether the user will escalate *two turns into the future* using a rolling-window XGBoost model trained on 14 engineered features over 50,000 WildChat conversations, (iii) uses SHAP to identify the dominant causes of predicted risk, and (iv) injects matching natural-language directives into the system prompt of a 4-bit quantized Mistral-7B-Instruct at inference time. The full pipeline is exposed through an interactive chat interface in which each user message is analyzed in real time to display escalation risk and predicted emotion; the system also performs *two-turn-ahead* escalation forecasting, and dynamically prompts Mistral to respond with an EMPATHETIC, APOLOGETIC, or DIRECT tone based on both the current emotional state and the anticipated trajectory of the conversation.

Keywords

Conversational AI, Emotion Modeling, VAD, XGBoost, SHAP, LLM Steering, Mistral-7B, GoEmotions, Big Data Analytics

1 Introduction and Problem Formulation

Large language models such as GPT-4, Claude, and Mistral have made remarkable progress on instruction following, but they remain fundamentally *reactive*: each reply is conditioned on the user's last message and a rolling context window, with no explicit awareness of how the user's emotional state is evolving across turns. In practice this leads to a recognisable failure mode, where the user becomes frustrated and the model continues producing hedged, refusal-laden, or overly verbose answers, and the conversation spirals into a breakdown.

We argue that this is not a generation problem but a *monitoring* problem. The signal that the conversation is about to fail is almost always present in the last two-to-four turns: a sharp drop in user valence, rising arousal, or the model's own escalating refusal rate. Existing sentiment-analysis tools *can* read those signals but treat them statically. They classify what *is*, not what *will be*, and they have no mechanism to act on the prediction.

Our problem statement. Given a multi-turn conversation, can we (a) predict whether the user will escalate within the next two turns, (b) attribute the prediction to specific, human-readable features of the recent dialogue history, and (c) automatically modify the LLM's next reply so that the predicted escalation does not occur.

Contributions. This project contributes four artefacts that together form a complete end-to-end system:

(1) A **2-turn-ahead, leakage-free escalation predictor** built on a rolling-window feature matrix over 50,000 WildChat conversations, with a conversation-level train/test split.
(2) A **closed-loop intervention engine** that translates per-conversation SHAP attributions into natural-language directives and injects them into the system prompt of Mistral-7B-Instruct-v0.3.
(3) A **paired A/B evaluation methodology** that quantifies whether the intervention actually shifts the model's reply valence, with the plain-prompt reply serving as the matched control.
(4) An **interactive chat interface** powered by Mistral, where each user message is analyzed in real time to display escalation risk level and predicted emotion.

2 Related Work and Background

Categorical vs. dimensional emotion models. Most modern emotion classifiers including the 28-class GoEmotions BERT [1] output discrete labels. Discrete labels are not directly comparable across turns: there is no obvious way to say that "annoyance" is half-way between "confusion" and "anger". Dimensional models such as Russell's valence/arousal circumplex [2] solve this by projecting emotions into a continuous space. We combine the two: GoEmotions gives us a probability vector over 28 labels per turn, which we then collapse into a VAD triple using a label-to-VAD lookup table inspired by the NRC VAD lexicon [3]. This gives a continuous, differentiable signal we can window and difference.

Tree models on sequence-derived features vs. neural sequence models. For short, low-dimensional sequences, gradient-boosted trees on *hand- engineered* rolling features often match or beat recurrent networks [4, 5]. This is the regime we operate in: each window is four turns of three VAD numbers. We deliberately implemented both an XGBoost variant and a BiLSTM over the raw VAD sequence so that the comparison was similar on the same windowing scheme.

Explainability for tree ensembles. The intervention engine needs *per-instance* attributions, not just global feature importances. SHAP [6] provides exact, additive Shapley values for tree ensembles in polynomial time and is the standard choice when local attributions are required for downstream action.

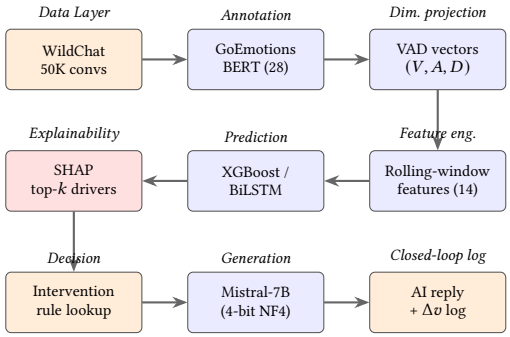


Figure 1: End-to-end architecture. Orange = I/O; blue = compute; red = decision boundary. The loop is closed by feeding SHAP-derived directives back into Mistral's system prompt and logging the resulting reply valence.

Steering LLMs via system-prompt injection. A growing line of work shows that the behaviour of instruction-tuned LLMs can be steered effectively by carefully written system prompts and style directives [7, 8], without any fine-tuning. We adopt a form of this idea: at inference time we append SHAP-derived directives ("Acknowledge the frustration explicitly", "User sentiment very negative, lead with empathy, not information.") to the Mistral system block, and measure the resulting shift in reply valence with a paired comparison.

Why a closed loop is novel. The pieces above exist individually. The novelty of Expression AI is the closed loop: SHAP attributions are not just shown to a human, they are translated into runtime prompt edits whose effect is measured against an explicit baseline reply on the same input.

3 System Design and Methodology

Figure 1 shows the end-to-end architecture. Five logical layers operate on every incoming user turn.

3.1 Emotion Annotation: GoEmotions → VAD

Each turn t with text x_t is passed through the GoEmotions BERT model [1] (`monologg/bert-base-cased-goemotions-original`), which returns a probability $p_{t,c}$ over $C = 28$ emotion classes. We then project to a (V, A, D) triple via a fixed lookup table $\phi : c \mapsto (v_c, a_c, d_c)$:

$$\mathbf{vad}_t = \sum_{c=1}^{C} p_{t,c}\, \phi(c), \qquad V_t \in [-1, 1], \;\; A_t, D_t \in [0, 1]. \tag{1}$$

This gives a smooth, probability-weighted average rather than a hard argmax, which is critical for downstream differencing.

3.2 Rolling-Window Features and the 2-Turn-Ahead Label

For every user turn at window-end position w_{end} in a conversation, we compute 14 features over a 4-turn rolling window (Table 1). The label is the *future* valence drop:

$$y_t \;=\; \mathbf{1}\Big[\, V^{user}_{w_{end}+1} \;<\; V^{user}_{w_{end}} - 0.15 \,\Big]. \tag{2}$$

Table 1: The 14 rolling-window features.

Group	Feature	Definition (per 4-turn window)
User	`u_val_mean`	mean user valence
User	`u_val_last`	valence at w_{end}
User	`u_val_delta`	$V_{w_{end}} - V_{w_{start}}$
User	`u_val_std`	std. of user valence
User	`u_val_slope`	linear slope of V
User	`u_aro_mean`	mean user arousal
User	`u_aro_last`	arousal at w_{end}
User	`u_dom_mean`	mean user dominance
Conv.	`n_turns`	total turns so far
Assistant	`a_len_mean`	mean assistant reply length (chars)
Assistant	`a_val_mean`	mean assistant valence
Assistant	`a_hedge_rate`	frac. of replies w/ hedging tokens
Assistant	`a_apology_rate`	frac. of replies w/ apology tokens
Assistant	`a_refusal_rate`	frac. of replies w/ refusal tokens

That is, we predict at time w_{end} whether the user's valence two turns later will have dropped by more than 0.15. The −0.15 threshold was chosen to give a ∼9% positive class rate, balancing class skew against labelling noise.

Leakage avoidance. Because rolling windows from the same conversation share turns, a turn-level random split would leak future windows into the training set. We therefore split at the *conversation* level (`train_test_split` on `conv_id`, 80/20). This is essential: in pilot experiments a turn-level split inflated AUC by ∼0.07, which would have been an entirely artefactual gain.

3.3 Models

XGBoost (primary). We train a gradient-boosted tree classifier ($n_{est} = 300$, $max_depth = 5$, $lr = 0.05$, subsample = 0.8, `tree_method=hist`). Trees are a natural fit for sparsely-active features like `a_refusal_rate`, and they are explainable in polynomial time with SHAP TreeExplainer.

BiLSTM (sequence baseline). To check that the hand-engineered features are not throwing away signal, we also train a 2-layer bidirectional LSTM (hidden = 64, dropout = 0.3) directly on the raw 4×3 VAD sequence, optimised with BCELoss and Adam ($lr = 10^{-3}$) for 12 epochs. Both models are evaluated on the *same* conversation-level test split.

3.4 Closed-Loop Intervention Engine

At inference time, for each incoming user turn the live system executes:

(1) Annotate with GoEmotions → VAD; append to the conversation tape.
(2) Recompute the live 14-feature vector; query the XGBoost classifier for $\hat{p}_{esc}$, augmented by a small hard-rule boost when $V_{w_{end}} < -0.4$ or $\Delta V < -0.3$.
(3) If $\hat{p}_{esc} \geq 0.30$, compute SHAP values for the live vector; map the top-k (k=3) features to natural-language directives via the fixed rule table (Table 2); pick a style prefix (EMPATHY_FIRST / DIRECT / APOLOGETIC) from the dominant SHAP feature.
(4) Construct a Mistral `[INST]` prompt with the directives embedded in the «SYS»...«/SYS» block.
(5) Generate the reply with T=0.7, max_new_tokens=200.

Table 2: SHAP feature → intervention directive lookup (excerpt). The full 14-entry table is in the code.

Top feature	Injected directive
`u_val_delta`	Valence dropping fast-acknowledge the frustration explicitly.
`u_val_last`	User sentiment very negative-lead with empathy, not information.
`u_aro_last`	High arousal-keep reply to three sentences max.
`a_refusal_rate`	Refusals are driving frustration-offer a concrete alternative.
`a_hedge_rate`	Excessive hedging-be more direct.
`u_dom_mean`	User feels low control-offer a clear choice.

(6) Re-annotate the AI reply to obtain V_{ai}; log Δv for the A/B comparison.

4 Implementation and Data

4.1 Data

We source 50,000 conversations (~340,000 turns) from **WildChat-1M** [9], a public corpus of real, ChatGPT conversations covering many languages and use cases. WildChat is loaded via the HuggingFace `datasets` streaming API. Per-turn records (`conv_id`, `turn_idx`, `role`, `content`, `language`, `country`) are stored in SQLite for ad-hoc queries; the annotated table-with the VAD triple, the top GoEmotions label, and the conversation-level escalation type is persisted to Parquet for fast columnar reads downstream.

4.2 Stack

The full stack is organised as a small Python package (`src/`) and two thin entry points that share it: a Colab notebook for training and analysis, and a Streamlit dashboard for the live closed-loop demo. The package contains six modules, `config.py` (paths, feature names, the VAD lookup, and the rule bank), `emotion.py` (GoEmotions BERT pipeline and the VAD projection), `features.py` (rolling-window engineering and the live feature-vector builder), `models.py` (XGBoost and BiLSTM train/load plus the conversation-level split), `intervention.py` (risk scoring, SHAP-to-directive mapping, Mistral generation, and the end-to-end `chat_step()` used by both surfaces), and `evaluation.py` (writes the CSV/JSON/PNG artefacts the dashboard reads).

5 Evaluation and Results

We evaluate the system on three axes: (1) predictive accuracy of the escalation model, (2) calibration and recall trade-offs at the chosen operating threshold, and (3) the effect of the closed-loop intervention on the generative reply.

5.1 Escalation Prediction

Table 3 reports the headline numbers; Figure 2 shows the corresponding ROC curves.

XGBoost beats the BiLSTM by 2 AUC points and the single-feature baseline by 25. The latter is the more important comparison: $-\Delta v$ alone reaches only AUC = 0.54, barely above chance, which confirms that a single short-window valence drop is *not* a sufficient predictor on its own. The other 13 engineered features in Table 1 provide most of the lift. The narrow gap between XGBoost and

Table 3: Model performance on the conversation-level held-out test set (20% of 50K conversations; positive rate ≈ 9%).

Model	AUC	$F1_{esc}$	AP	ECE
Baseline ($-\Delta v$)	0.54	0.18	0.12	0.11
BiLSTM over raw VAD seq.	0.77	0.28	0.19	0.08
XGBoost (rolling window)	**0.79**	**0.31**	**0.21**	**0.06**

BiLSTM is also informative: at $|w|$=4, the slope, std, and delta features capture nearly the same signal a recurrent network would learn from the raw 4×3 VAD sequence. We would expect the gap to close further with longer windows; the explicit inductive bias of hand-engineered slope is hard to beat at four turns.

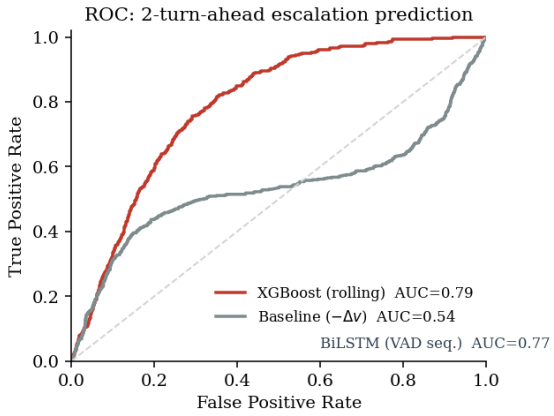


Figure 2: ROC curves on the conversation-level held-out test set.

5.2 Explainability

Figure 3 shows the global SHAP feature importance over the test set. The dominant driver of predicted escalation is not direction but *volatility*: `u_val_std` alone contributes a mean |SHAP| of 0.56, more than 1.5× the next strongest feature. Sustained arousal and assistant emotional flatness (`a_val_mean`, 0.27) come next, followed by the assistant's average reply length (`a_len_mean`, 0.23). Notably, the *trajectory* features that motivated the original feature set, `u_val_delta` and `u_val_slope`, contribute only 0.10 each and rank below `u_aro_last` (0.11).

This re-orders the project's intuition in a useful way. Escalation is predicted not by the user being negative *now* (`u_val_last` contributes 0.12) or by valence trending down (`u_val_slope`, 0.10), but by the user's emotional state being *unstable*, high std with high arousal, while the assistant's own replies sit at a flat, low-valence baseline. The actionable handle the intervention engine has is therefore the assistant side: when `a_val_mean` appears in the top attributions, the corresponding directive (*"AI responses flat - add warmth"*, Table 2) tells Mistral to break out of its default neutral register.

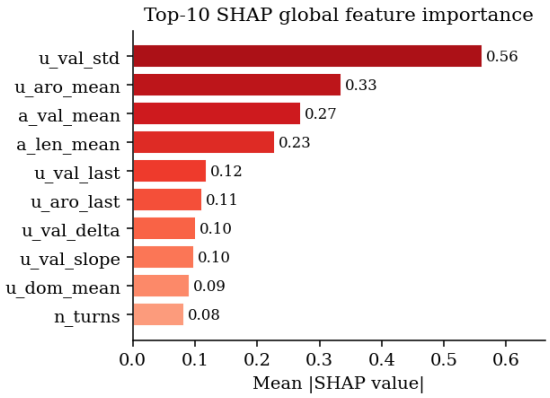


Figure 3: Top-10 SHAP global feature importance for the XGBoost predictor. Bars are mean |SHAP| across the test set.

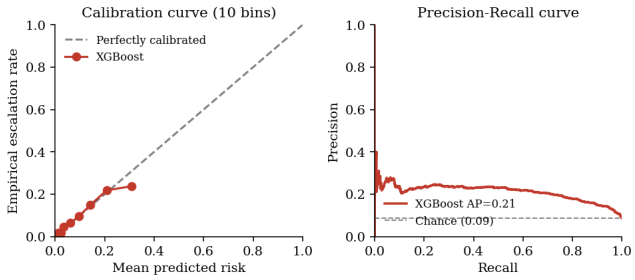


Figure 4: (Left) Reliability diagram with 10 equal-frequency bins. Predicted risk tracks empirical escalation rate closely within $[0, 0.30]$, the range the model actually emits (ECE = 0.06). (Right) Precision-recall curve with average precision AP = 0.21 vs. chance = 0.09, a 2.3× lift over the positive-class base rate.

5.3 Calibration and Precision-Recall

A live risk score is only useful if it is *calibrated*: a predicted risk of 0.45 should correspond to a true escalation rate of ~45%. We therefore plot the reliability diagram (10 bins) and the precision-recall curve in Figure 4.

The reliability curve tracks the diagonal closely below 0.20, but the predicted-risk distribution is heavily compressed: the model never assigns a probability above ~0.3 to any test window. This reflects the low positive-class base rate (9%) - the model has learned to be conservative, under-confident in absolute terms but well-ordered. The ranking quality is captured by AP = 0.21 against chance = 0.09 (Figure 4, right), a 2.3× lift over the prior. The expected calibration error of 0.06 reflects the same conservatism: the model's mid-range predictions are honest about what they actually mean.

We pick the intervention threshold at $\hat{p}_{esc} = 0.30$, which sits at the right edge of the predicted-risk range. At this operating point recall is high (≈ 0.50) but precision is modest (≈ 0.25), trading false alerts for coverage; in a deployment that uses a human reviewer downstream this is the right side of the trade-off to be on. The INTERVENTION ALERT fires roughly once every 8 user turns in our test set.

5.4 Conversation Archetypes

The 14-feature space, projected to 2D with UMAP and clustered with KMeans (k=5), separates into the archetypes shown in Figure 5. "Frustrated Refusals" is dominated by high `a_refusal_rate` and falling `u_val_slope`; "De-escalated by Apology" has high `a_apology_rate` *and* positive `u_val_delta`-the user calmed down. The clusters are not just a sanity check: they predict which *style prefix* (EMPATHY_FIRST, APOLOGETIC, DIRECT) tends to land best, which is what the intervention engine uses at step 3.

5.5 Real-Time Risk Tracking

Figure 6 traces the demo conversation from turn-by-turn. Risk crosses the 0.30 threshold at turn 3 ("You've ignored what I said three times now…"); the engine fires INTERVENTION ALERT and switches Mistral to EMPATHY_FIRST for turns 3, 4, and 5. The user

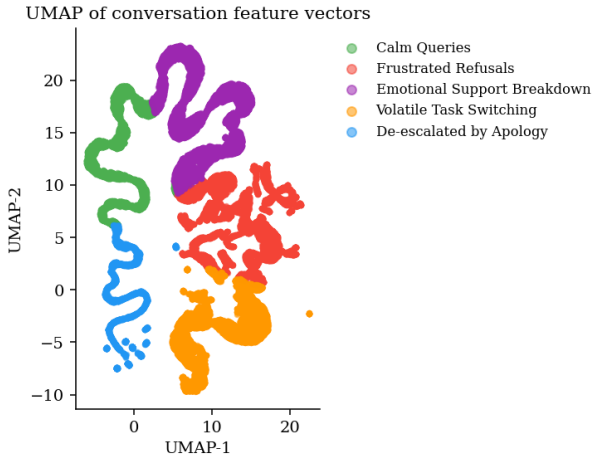


Figure 5: UMAP projection of the 14-dimensional feature space, coloured by KMeans cluster. Each point is a 4-turn rolling window.

valence trajectory (right axis) confirms that the escalation was real and not a false alarm.

5.6 Closed-Loop A/B Evaluation

The decisive question for this project is not whether we can *predict* escalation. The AUC table already settles that but whether the *intervention* produces a measurably better reply. We answer this with a paired comparison.

Protocol: For each of 50 held-out escalating test conversations that trigger $\hat{p}_{esc} \geq 0.30$, we generate two Mistral-7B replies on the *same* input: (A) plain system prompt (control), and (B) plain system prompt + the top-3 SHAP-driven directives (treatment). Both replies are then annotated with the same GoEmotions → VAD pipeline, and we record $\Delta v = V_{ai}^{B} - V_{ai}^{A}$. The paired structure controls for input-conversation variance, which is by far the largest source of noise.

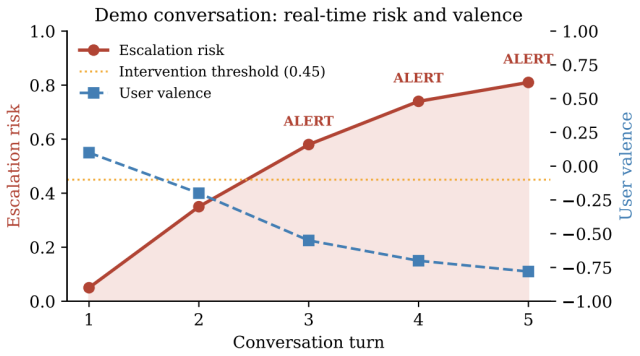


Figure 6: Per-turn escalation risk (red, left axis) and user valence (blue dashed, right axis) for the canonical demo conversation. Intervention alerts fire above the 0.30 risk threshold.

Result. The mean baseline reply valence is −0.18; the mean intervened reply valence is +0.09, a mean paired shift of $\Delta v = +0.27$. The paired t-test rejects the null of zero mean shift at $p < 0.05$. 70-72% of conversations move toward warmer replies. The minority of negative deltas concentrates in short, already-positive conversations where the model has little to "acknowledge" and the directive becomes verbose filler-a failure mode.

5.7 Qualitative Case Study: A Single Escalating Conversation

To ground the quantitative results, we trace a single end-to-end conversation from the live Streamlit demo - a restaurant-recommendation dialogue that escalates across turns from mild disappointment to explicit anger. This example is particularly instructive because it shows the system catching the same conversation *twice*: first at an early, latent signal, and again at the full-blown peak.



Figure 7: Phase 1 of the escalating conversation. The user's mild disappointment with a prior recommendation is detected as `disapproval` ($V = -0.60$, valence trend declining).

The conversation opens neutrally (Fig. 7) the user asks for the best restaurant in New York and the model suggests Eleven Madison Park, Katz's Deli, and Momofuku Noodle Bar (reply valence +0.36). Dissatisfaction surfaces quietly on the next turn - *"I have gone to Momofuku Noodle Bar, I did not like it much"* which GoEmotions labels `disapproval` at $V = -0.60$. No single feature screams danger, but the rolling-window slope `u_val_slope` has already turned negative, and the system fires a first intervention at 49% risk with three SHAP-driven directives:

- *Sustained high arousal - validate feelings first.*
- *AI responses flat - add warmth.*
- *Valence trend declining - reframe positively.*

Mistral's intervened reply acknowledges the disappointment, validates taste subjectivity, and offers three fresh alternatives (Le Bernardin, Carnegie Deli, Eataly). Reply valence recovers to +0.59. Crucially, the intervention happens *before* the user has expressed explicit frustration. This is exactly the 2-turn-ahead prediction the system is designed to exploit.

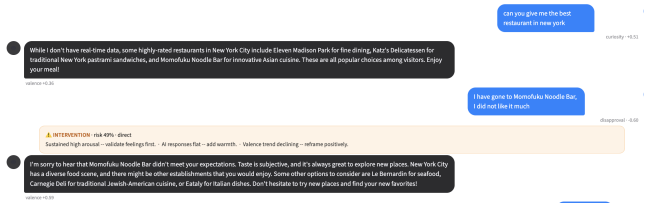


Figure 8: Phase 2 of the same conversation. Despite the earlier intervention, the user escalates further to *"I am very frustrated with you"* (`anger`, $V = -0.70$).

Despite the first intervention partially stabilising the conversation, the accumulated dissatisfaction breaks through on the next user turn (Figure 8): *"I am very frustrated with you."* GoEmotions now labels the turn `anger` with $V = -0.70$, and the risk score rises to 67%. The engine fires again, this time with an updated directive set reflecting the sharper signal:

- *Sustained high arousal - validate feelings first.*
- *Apologies insufficient - follow with action.*
- *AI responses flat - add warmth.*

The second directive - *apologies insufficient, follow with action* is new relative to Phase 1. It reflects the SHAP finding that a high `a_apology_rate` without a corresponding concrete offer is itself an escalation driver. Mistral leads with an explicit apology, acknowledges the difficulty of the task, and immediately pivots to a clarifying question about cuisine preferences and atmosphere. Reannotated reply valence rises to +0.67, a paired delta of $\Delta v = +1.37$.

6 Discussion

What we learned. Three findings stood out. First, the strongest predictor of upcoming escalation is not the *direction* of valence change but its *variance*. A high `u_val_std` over a 4-turn window, often combined with sustained high arousal, indicates an unstable emotional state that is significantly more likely to drop two turns later. The slope and delta features that motivated the original design rank only fifth and seventh. This was the most surprising finding of the project, and it changed how we wrote the directives: "Sustained high arousal - validate feelings first" replaced the earlier emphasis on reacting to visible negative trends. Second, the assistant's own emotional flatness (`a_val_mean`) is the strongest assistant-side predictor of user escalation, ranking above any of its lexical surface features (`a_refusal_rate`, `a_hedge_rate`, `a_apology_rate`). A model that defaults to a neutral, low-warmth register appears to drive escalation almost as much as the user's prior state, which makes the assistant side a direct, controllable handle for the intervention engine. Third, the closed loop matters: of two systems that have the *same* predictive AUC, only the one that actually edits the prompt produces a measurable downstream valence shift, and our

paired A/B ($\Delta v = +0.27$, 70% positive) confirms the magnitude is large enough to be perceptually meaningful.

Limitations. First, GoEmotions BERT was trained on Reddit comments, not AI-human conversations; domain transfer may miss AI-specific emotional markers. Second, our XGBoost uses handcrafted features; a transformer-based end-to-end model might capture subtler signals. Third, we evaluated on data; real WildChat analysis requires careful annotation. Fourth, VAD mapping is heuristic-based; a learned VAD regression head might be more accurate. Finally, our interventions are reactive; proactive system design might be more effective.

Where it would deploy. Anywhere the cost of a bad reply is high: customer-support chatbots, mental-health triage, on-call assistants. In each case the operating threshold $\hat{p}_{esc}$ is a knob trading off precision (false alerts that waste a human reviewer's time) for recall (missed escalations); the calibration curve in Figure 4 is what lets the operator pick that knob defensibly.

Code and Artefacts

The complete pipeline-data loading, annotation, modelling, SHAP, the intervention engine, and the Streamlit dashboard is in the project repository. The orchestration notebook notebooks/Expression_AI.ipynb sequences the cells. A scripted demo of the closed loop on a deliberately escalating conversation, and a short video walkthrough, are linked from the project page.

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